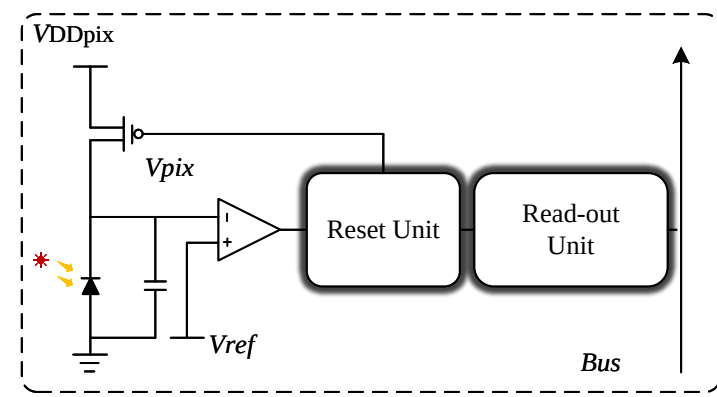
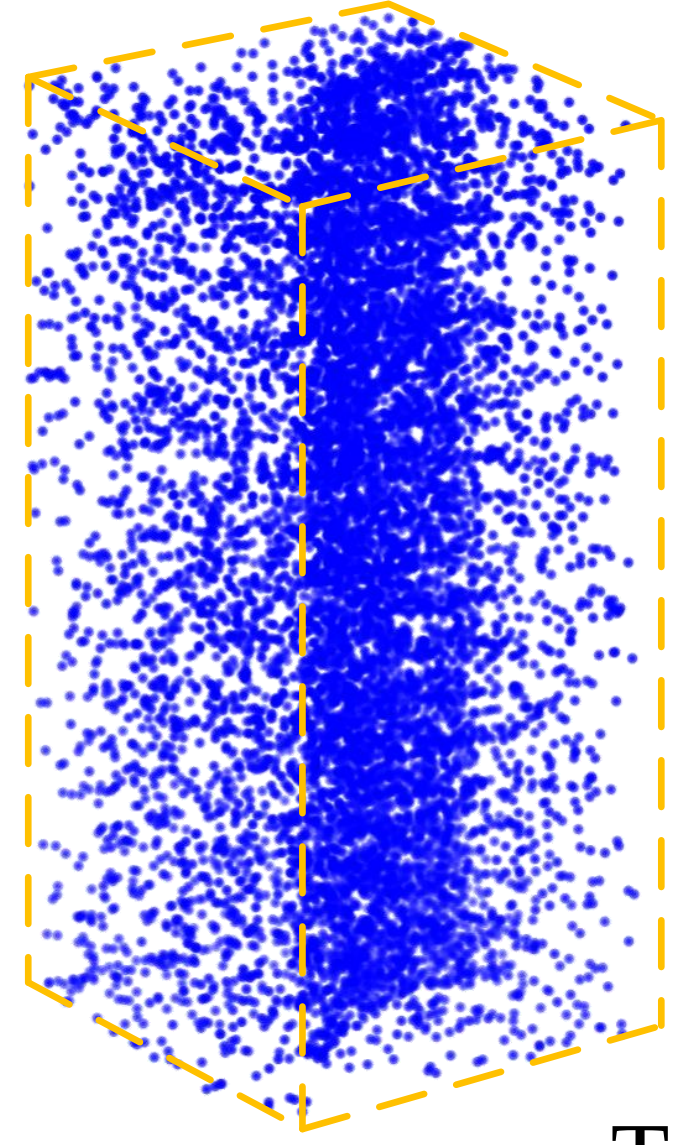


Dynamic Scene

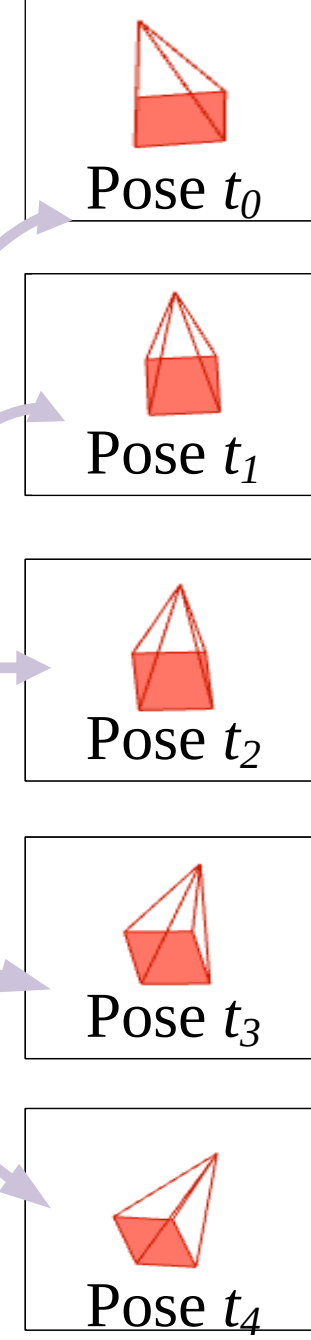


Dynamic Spike Stream



Spike Camera

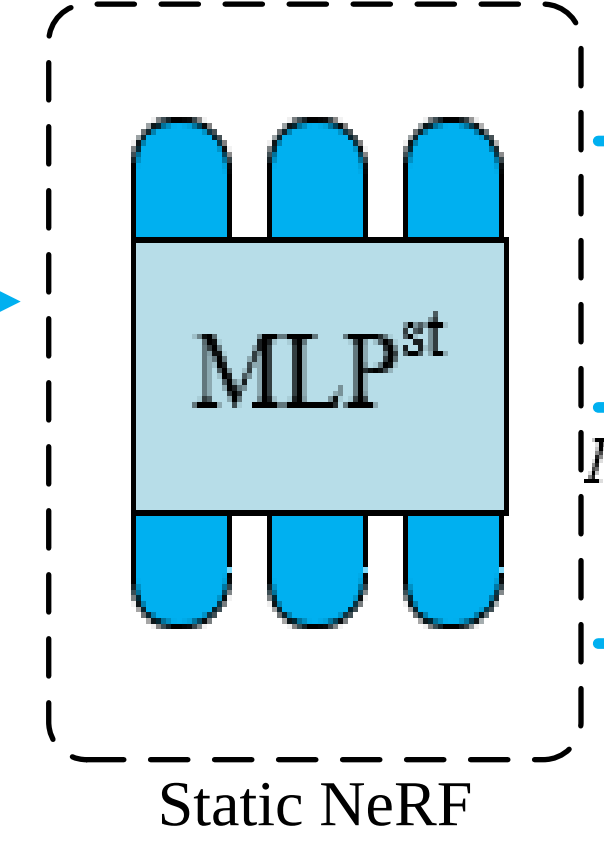
Pose Estimation Block



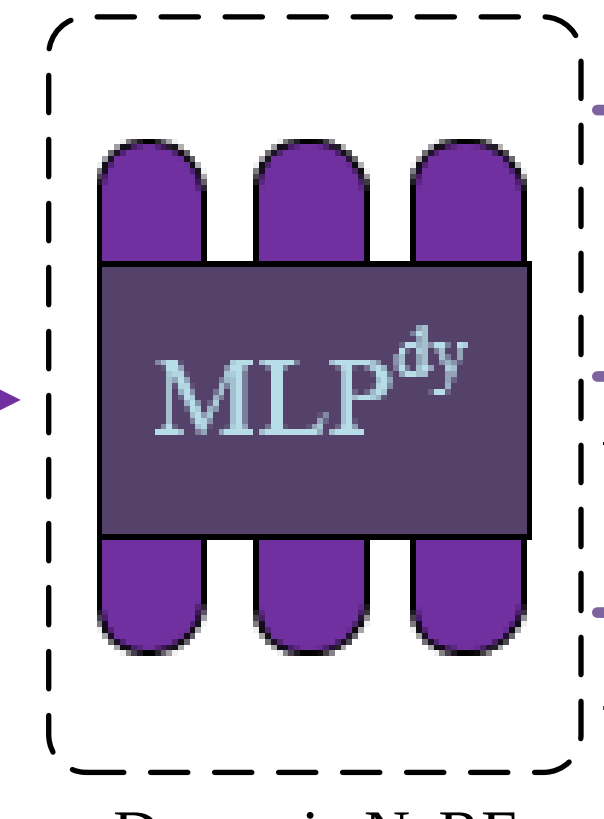
...

 t_i

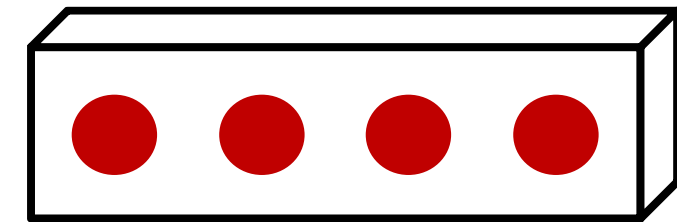
Volume Rendering



Static NeRF

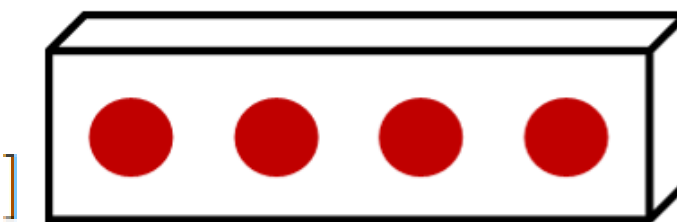


Dynamic NeRF



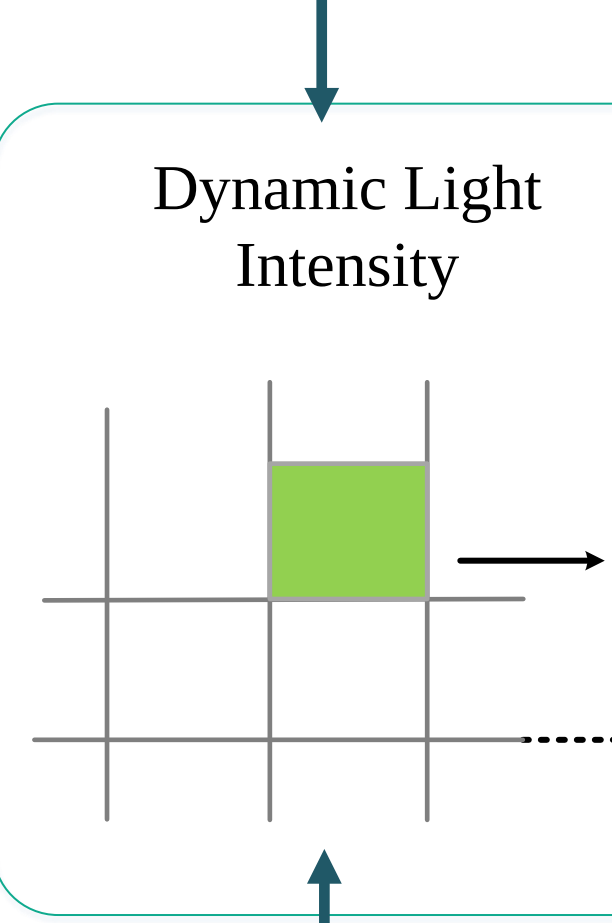
$$\mathcal{L}_{cb} = \sum_{r \in \mathbb{R}} [\|D(r, S)\|_2^2]$$

$$\mathcal{L}_{pho} = \sum_{r_i \in \mathbb{R}} \sum_{j \in \mathcal{N}(i)} [w_{j \rightarrow i}(r_i) \|D_{j \rightarrow i}(r_i, S)\|_2^2]$$



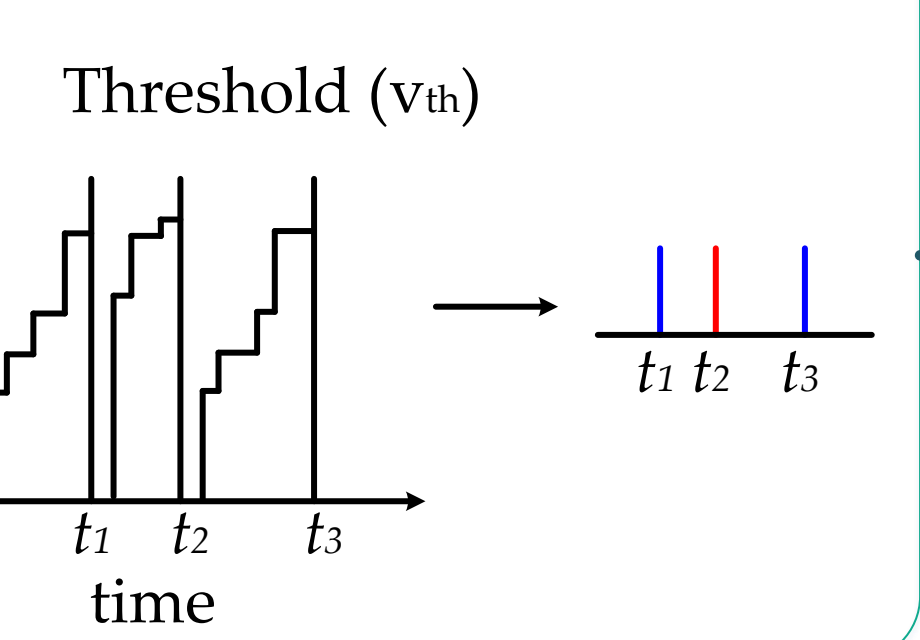
Intensity and Depth Blend

Intensity Correction



Dynamic Light Intensity

Integrate and Fire (IF) neuron



Spiking Neuron Layer

Depth, Flow and Loss

$$\mathcal{L}_{flow} = \|F(r) - F_{gt}(r)\|_1 \quad \mathcal{L}_{depth} = \|D(r) - D_{gt}(r)\|_1$$

$$\mathcal{L}_{cyc} = \sum w_{i \rightarrow i+1} \|\mathbf{f}_{i \rightarrow i+1}(\mathbf{x}_i) + \mathbf{f}_{i+1 \rightarrow i}(\mathbf{x}_{i \rightarrow i+1})\|_1$$

$$+ w_{i \rightarrow i-1} \|\mathbf{f}_{i \rightarrow i-1}(\mathbf{x}_i) + \mathbf{f}_{i-1 \rightarrow i}(\mathbf{x}_{i \rightarrow i-1})\|_1$$

$$D(r, S) = \sum_t \text{SN}(I(C_i(r))) \cdot R(x, y) - \sum_t I(C_i(F(S)))$$

Generated Spike Stream

